

Formula 1 Pit Stop Prediction Using Machine Learning

Kaggle Playground Series 2026

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Chapter 1

Introduction

Formula 1 racing is one of the most technologically advanced sports in the world. During a race, teams continuously make strategic decisions regarding tyre management, fuel consumption, weather conditions, driver performance, and pit stop timing.

Pit stops are critical because they directly influence race outcomes. A poorly timed pit stop can result in loss of track position, while an optimized strategy can improve race performance significantly.

This project focuses on predicting whether a Formula 1 driver will pit on the next lap using machine learning techniques. The dataset is provided as part of the Kaggle Playground Series 2026 competition and is inspired by real-world Formula 1 strategy data.

The primary objective is to build a robust predictive model capable of estimating the probability of a pit stop in the next lap using historical race and strategy-related features.

Chapter 2

Problem Statement

The objective of this project is to develop a machine learning model capable of predicting whether a Formula 1 driver will make a pit stop on the next lap.

The target variable is:

PitNextLap

where:

- 1 = Driver will pit on the next lap
- 0 = Driver will continue racing

The challenge involves analyzing racing patterns, tyre degradation, lap performance, and race strategy to accurately predict pit stop decisions.

The evaluation metric used in the competition is:

ROC-AUC Score

which measures the ability of the model to distinguish between positive and negative classes.

Chapter 3

Objectives

The major objectives of this project are:

- To understand Formula 1 race strategy data.
- To perform Exploratory Data Analysis (EDA).
- To preprocess and clean the dataset.
- To engineer meaningful racing and strategy-related features.
- To build machine learning classification models.
- To optimize the ROC-AUC score.
- To compare multiple machine learning algorithms.
- To generate predictions for unseen test data.

Chapter 4

Dataset Description

The dataset is synthetically generated but inspired by real Formula 1 race strategy data.

The dataset files include:

- train.csv
- test.csv
- sample_submission.csv

4.1 Training Dataset

The training dataset contains:

- Input features related to race conditions
- Target column: PitNextLap

4.2 Testing Dataset

The testing dataset contains all input features except the target variable.

4.3 Target Variable

Value	Meaning
0	No Pit Stop
1	Pit Stop on Next Lap

Table 4.1: Target Variable Description

Chapter 5

Exploratory Data Analysis

Exploratory Data Analysis (EDA) is performed to understand the structure and patterns of the dataset.

5.1 EDA Objectives

- Understand feature distributions
- Identify missing values
- Detect outliers
- Analyze correlations
- Study pit stop patterns
- Analyze class imbalance

5.2 EDA Techniques

- Histogram Analysis
- Boxplots
- Correlation Heatmaps
- Countplots
- Distribution Curves
- Feature Relationship Analysis

5.3 Important Insights to Explore

- Tyre age vs pit probability
- Lap number vs pit decision

- Driver behavior patterns
- Position changes
- Speed fluctuations

Chapter 6

Data Preprocessing

Data preprocessing is essential before model training.

6.1 Steps Involved

- Handling missing values
- Removing duplicate records
- Encoding categorical variables
- Feature scaling
- Data type optimization

6.2 Encoding Techniques

Categorical features are transformed using:

- Label Encoding
- Frequency Encoding
- Target Encoding

Chapter 7

Feature Engineering

Feature Engineering is the most important stage of this project.

7.1 Purpose

The dataset intentionally removes certain important information, therefore advanced feature engineering is required to extract hidden patterns.

7.2 Feature Engineering Techniques

7.2.1 Tyre Related Features

- Tyre Age
- Stint Length
- Tyre Wear Estimation
- Tyre Compound Encoding

7.2.2 Rolling Features

- Rolling Mean Speed
- Rolling Average Lap Time
- Rolling Position Changes

7.2.3 Lag Features

- Previous Lap Speed
- Previous Position
- Previous Sector Timing

7.2.4 Cumulative Features

- Total Distance Covered
- Cumulative Pit Count
- Average Pace

7.2.5 Race Strategy Features

- Gap to Leader
- Gap to Car Ahead
- Position Gain/Loss
- Fuel Load Approximation

Chapter 8

Machine Learning Models

Several machine learning models are used and compared.

8.1 Models Used

- Logistic Regression
- Random Forest
- XGBoost
- LightGBM
- CatBoost

8.2 Why CatBoost?

CatBoost is highly effective for tabular datasets because:

- Handles categorical variables efficiently
- Requires minimal preprocessing
- Reduces overfitting
- Provides strong ROC-AUC performance

Chapter 9

Model Evaluation

9.1 Evaluation Metric

The competition uses ROC-AUC score.

9.2 ROC-AUC Formula

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9.3 Performance Metrics

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

Chapter 10

Model Optimization

10.1 Hyperparameter Tuning

Optimization techniques include:

- Grid Search
- Random Search
- Bayesian Optimization

10.2 Cross Validation

Stratified K-Fold Cross Validation is used to improve generalization performance.

Chapter 11

Expected Outcomes

The expected outcomes of this project are:

- Accurate prediction of pit stop decisions
- Improved understanding of race strategy patterns
- High ROC-AUC score
- Strong feature engineering pipeline

Chapter 12

Future Enhancements

Future improvements may include:

- Deep Learning Models
- Sequential Time-Series Models
- LSTM-based Race Prediction
- Reinforcement Learning for Strategy Optimization
- Real-time Race Analytics

Chapter 13

Conclusion

This project demonstrates the application of machine learning techniques in Formula 1 race strategy prediction. By combining exploratory data analysis, feature engineering, and advanced ensemble models, it is possible to accurately predict pit stop decisions.

The competition provides an excellent opportunity to develop practical skills in tabular machine learning, feature engineering, and predictive analytics using real-world inspired racing data.

Chapter 14

References

- [Kaggle Playground Series 2026](#)
- [Formula 1 Strategy Analytics](#)
- [Scikit-Learn Documentation](#)
- [XGBoost Documentation](#)
- [LightGBM Documentation](#)
- [CatBoost Documentation](#)